Assignment 1 – Report

Module: Crime Mapping and Spatial Analysis - SECU0022

Candidate Number: XXXXX

Word Count: 1431

Date Sent: 14/February/2024

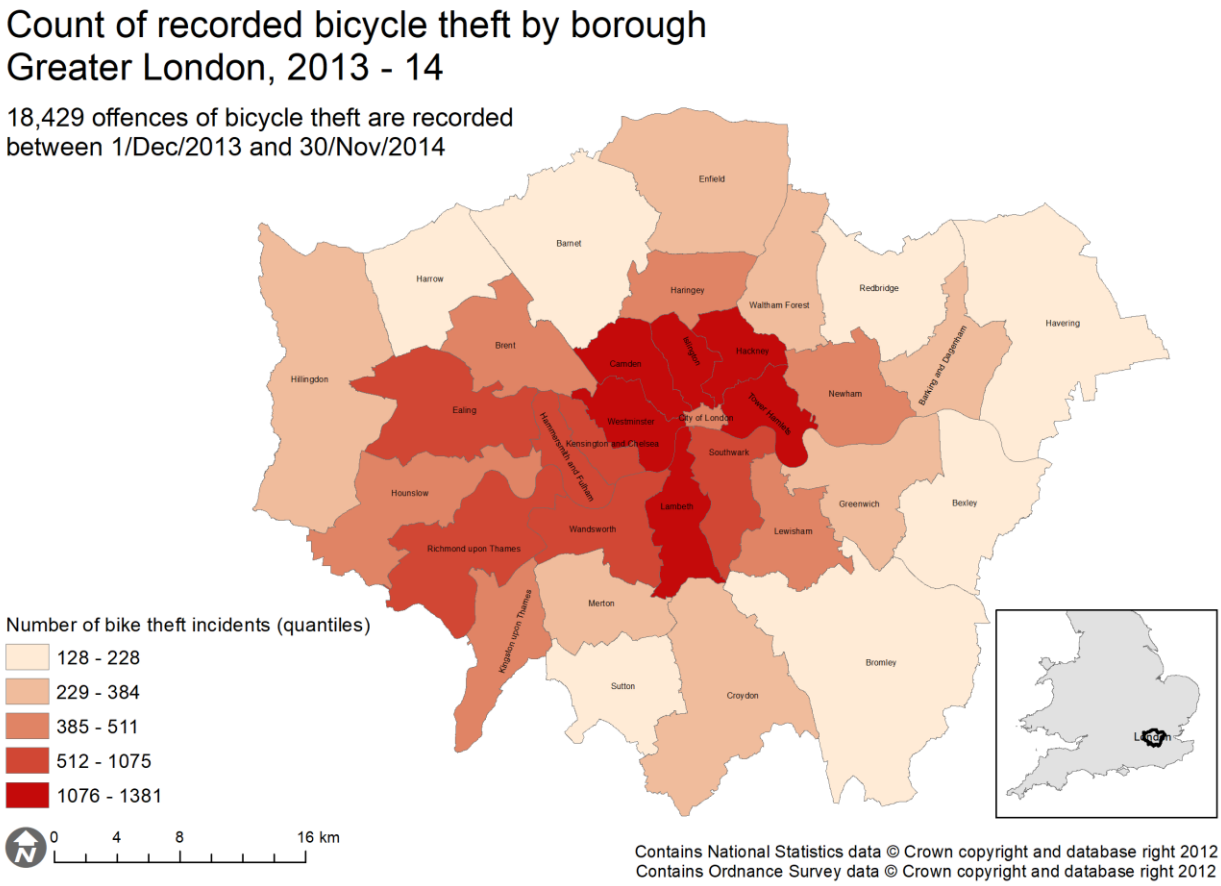
Introduction

Bicycle has becoming more popular as a medium of transportation in many countries, according to Transport of London (2007), bicycle use has increased by 83 percent since 2000. It is heathy and environmentally friendly, thus many companies encouraged their employees to cycle to work by giving out subsidies. However, these may not be effective towards objectives because they increase opportunities for bike theft.

Q1. Where, why

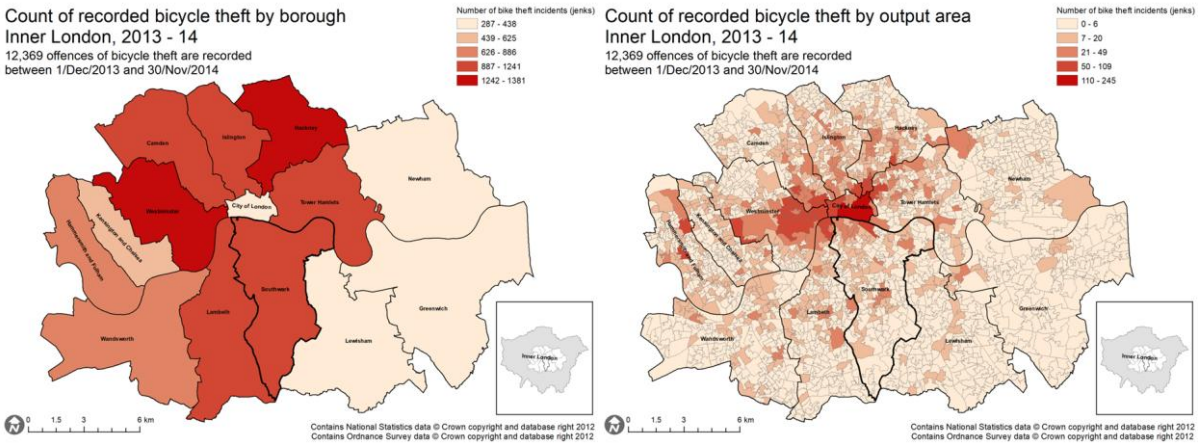
Recorded bike theft incidents are counted in borough level across Greater London as shown in Figure 1, darker shades of red illustrates higher number of incidents. Map shows crime is not distributed evenly, they concentrated in boroughs within inner part of London.

Figure 1. Bike theft incidents in Greater London by borough level



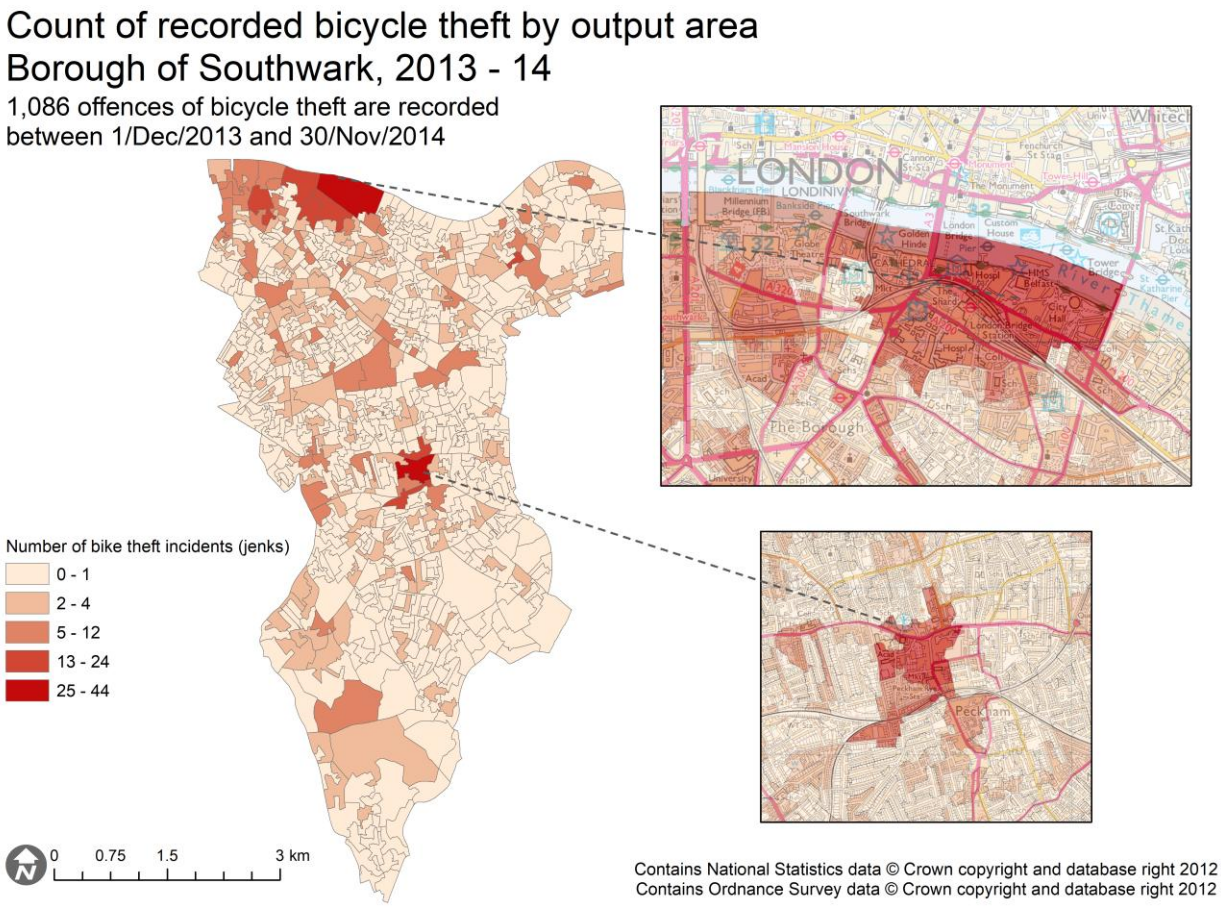
Zoom into Inner London, 12,369 incidents are recorded which account for 67% of the total incidents recorded across London. From Figure 2(a), it looks like boroughs around the City of London have the highest incidents, but using smaller geographic level in Figure 2(b), we could observe crime hotspots actually fall within the City of London. Ecological fallacy, also known as aggregation bias could be concluded. Although most incidents happened within the City of London, but it is not shown when aggregating the map to borough level as City of London has relatively small size. The borough of Southwark is highlighted in black border.

Figure 2. Bike theft incidents in Inner London by (a). borough (b). output area



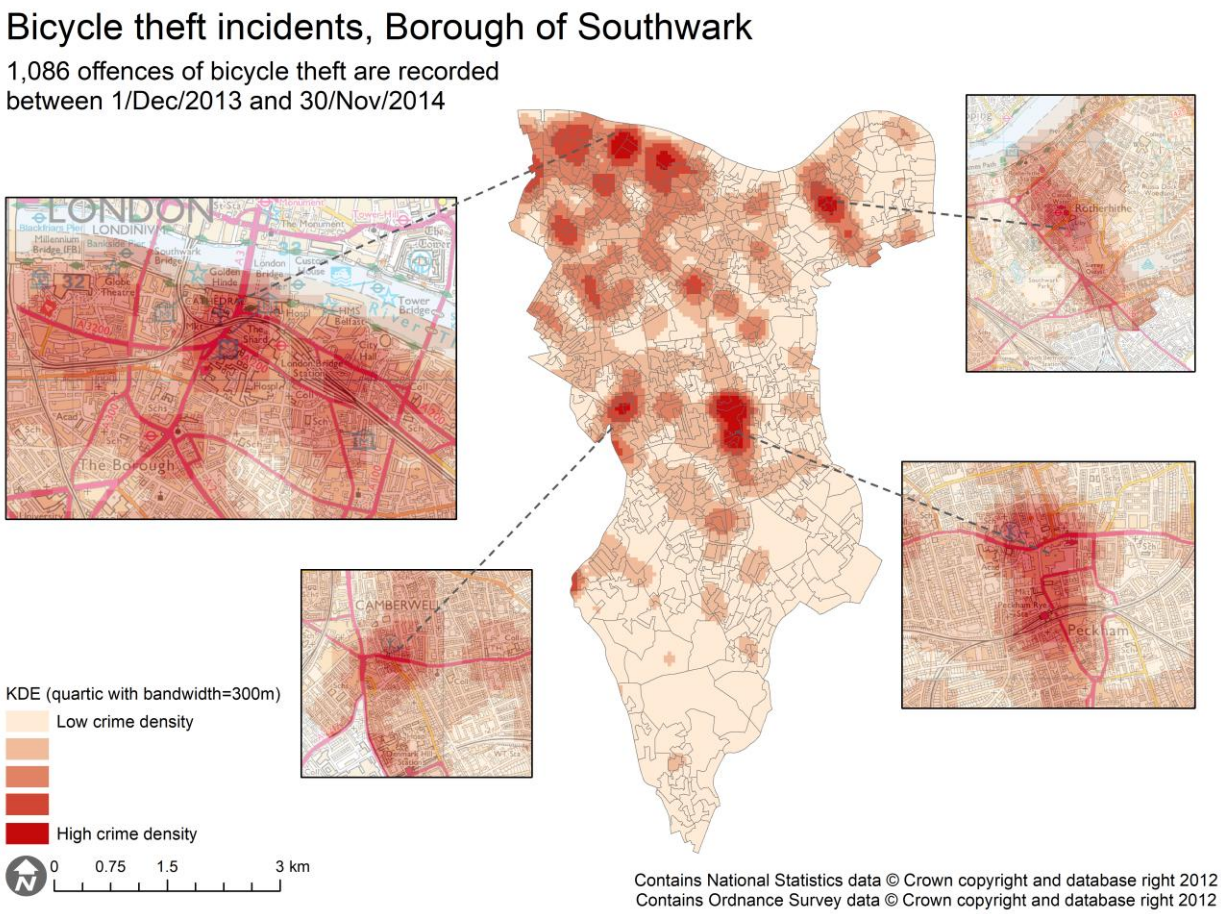
In examining the Southwark area by output area shown in Figure 3, we could observe two main hotspots where incidents are concentrated and shaded in dark red: one at the upper edge near the City of London and another located in the middle. When we put a closer look at these two areas with a detailed 1:25000 raster map, it becomes clear that these two spots possess specific land-use features that attract visitors. The upper spot has tourist attractions like Golden Hinge and HMS Belfast labelled as "☆". Also, there are museums labelled "🏛" and transport zones like piers and underground stations nearby. Another spot contains a leisure centre labelled "🏋", and a large library falls next to it. These suggest that these locations might have a connection to the crime we are interested in.

Figure 3. Bike theft incidents in Southwark by output area



By doing nearest neighbour analysis ,we obtain NNI=0.47, which means the average distance to the nearest neighbour is less than what would be expected from random distribution, with $p<0.001$, indicating crimes are significantly spatial clustered. As seen in Figure 4, the Kernal-density estimation map reveals crime hotspots more vividly. It highlights additional hotspots compared to the thematic map. This enhanced visibility because crimes often extend beyond administrative edges, and when data is aggregated by Output Areas (OA), it may not accurately reflect crime clusters that are close to those borders. It further reinforces the earlier point, bike theft incidents happened non-randomly and concentrate on featured places.

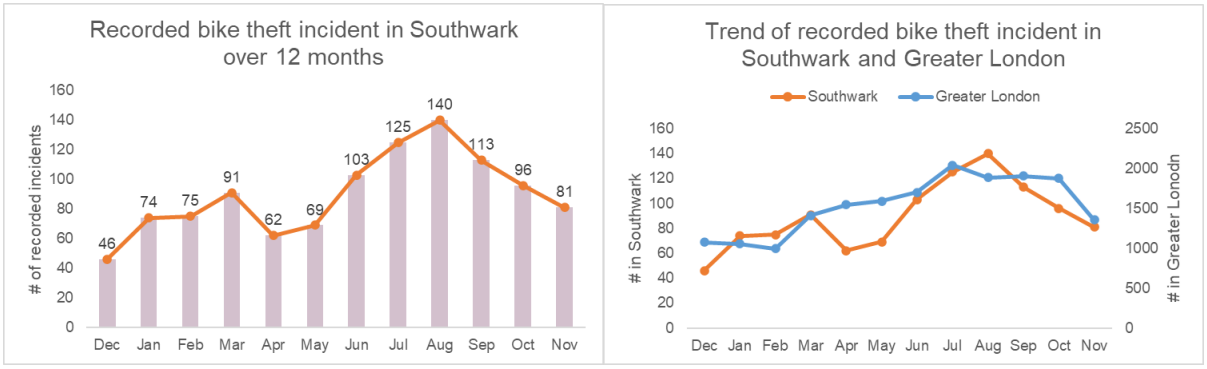
Figure 4. Kernal-density estimation map of Southwark



Q2. Crime frequency over time

The frequency of bike theft vary each month, as Figure 5 (a) shows, number of incidents for Southwark gradually increases from December to August, it decreases in April and May and reaches the peak with 140 incidents in August, then it starts to decrease in the winter. Similar trend can be observed in the Greater London from (b), where incidents are less in the winter and more in the summer, as less people would cycle when weather is colder, thus less opportunity for offenders in the winter.

Figure 5. Frequency of bicycle theft each month



(a) Southwark **(b) Southwark and Greater London**

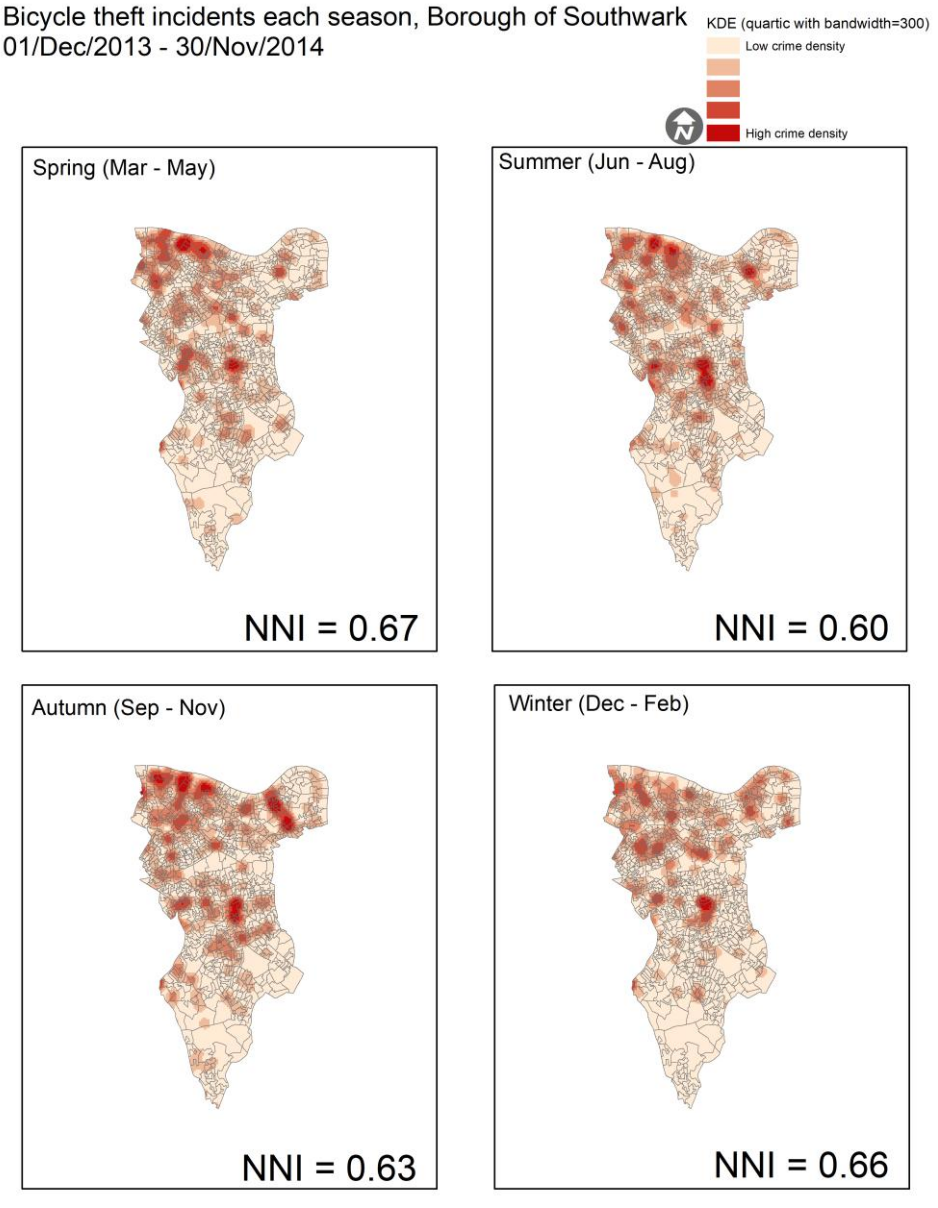
Comparing Figure 4 with Figure 6, crime hotspots we spotted earlier do not stay consistent throughout the year, there are less bike theft incidents near the top left tourist attraction area in December, instead crimes concentrated more near tube stations.

Figure 6. Kernal-density estimation maps of Southwark by month



In Figure 7, we have a seasonal hotspots map that includes the Nearest Neighbour Index (NNI) for each season, which helps to show if the crimes tend to be spatial clustered. The NNI is below 1 for all seasons, suggesting that crimes are indeed gathered together closely in space. However, it's noteworthy that during the summer, the spatial clustering of crimes is more evident compared to the winter. Furthermore, through examining the four maps and pinpointing the darkest areas, it's observable that various seasons exhibit distinct primary hotspots, with autumn showcasing particularly noticeable hotspots near the top right section. Recognizing that hotspots vary with the seasons, it becomes possible to allocate resources more effectively.

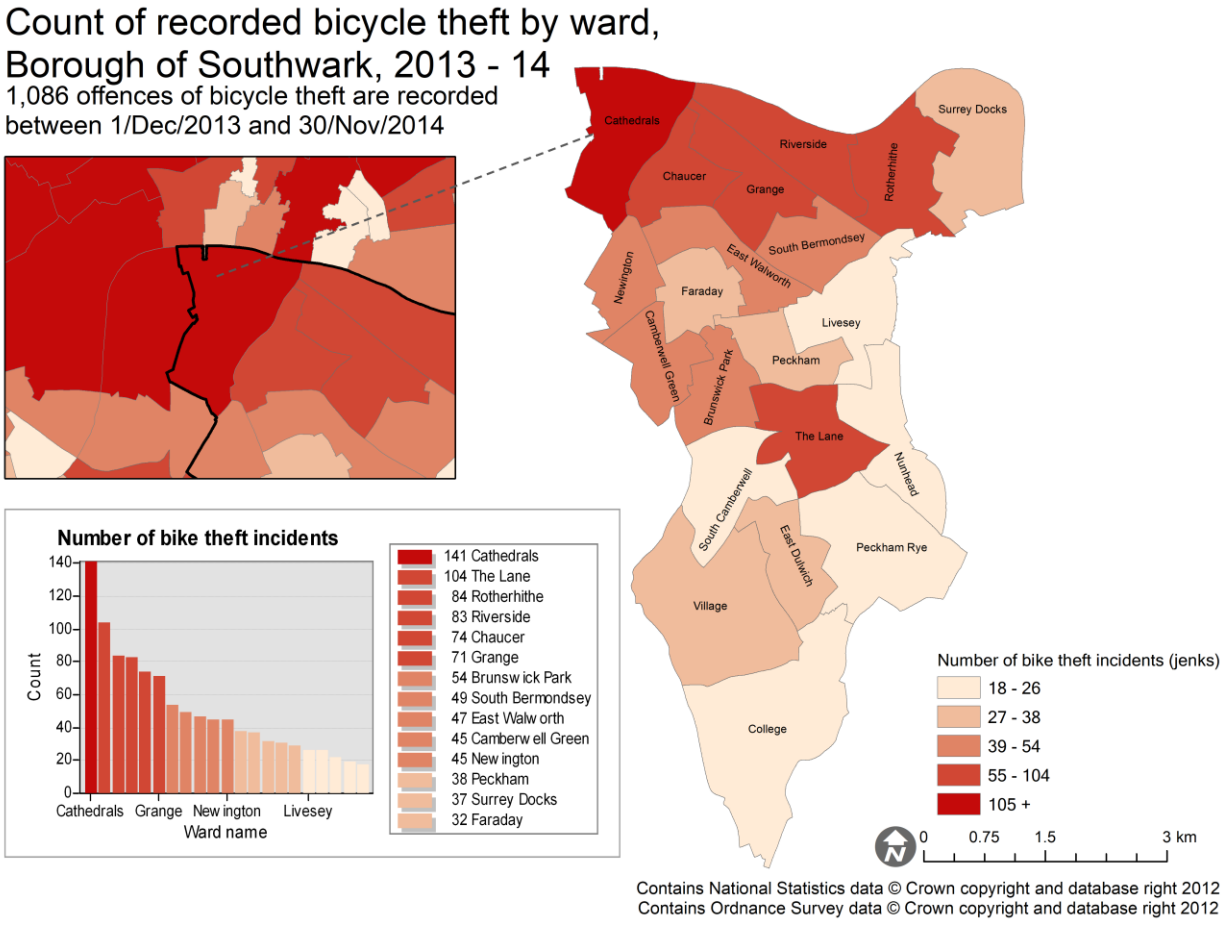
Figure 7. Kernal-density estimation maps of Southwark by season



Q3. Funding to a ward

By aggregating bike theft incidents to ward level as shown in Figure 8, we see that Cathedrals records the most crimes of 141 cases, the second place is The lane with 104 incidents recorded. Highest crime count could be explained by its proximity to other high-risk areas at the borough's edge as the inset map shows, more resources could be deployed here, funding police team in Cathedrals would be sensible since the ward has highest volume of recorded crimes. However, when we consider the crime rates adjusted for the at-risk population within each ward as shown in Figure 9, Cathedrals appears less risky because it has a larger population, resulting in a lower crime rate after adjustment. Despite this, focusing on area with the highest crime volume, such as Cathedrals, could still be a prudent strategy in terms of resource allocation.

Figure 8. Bike theft incidents in Southwark by ward



Q4. Risky place

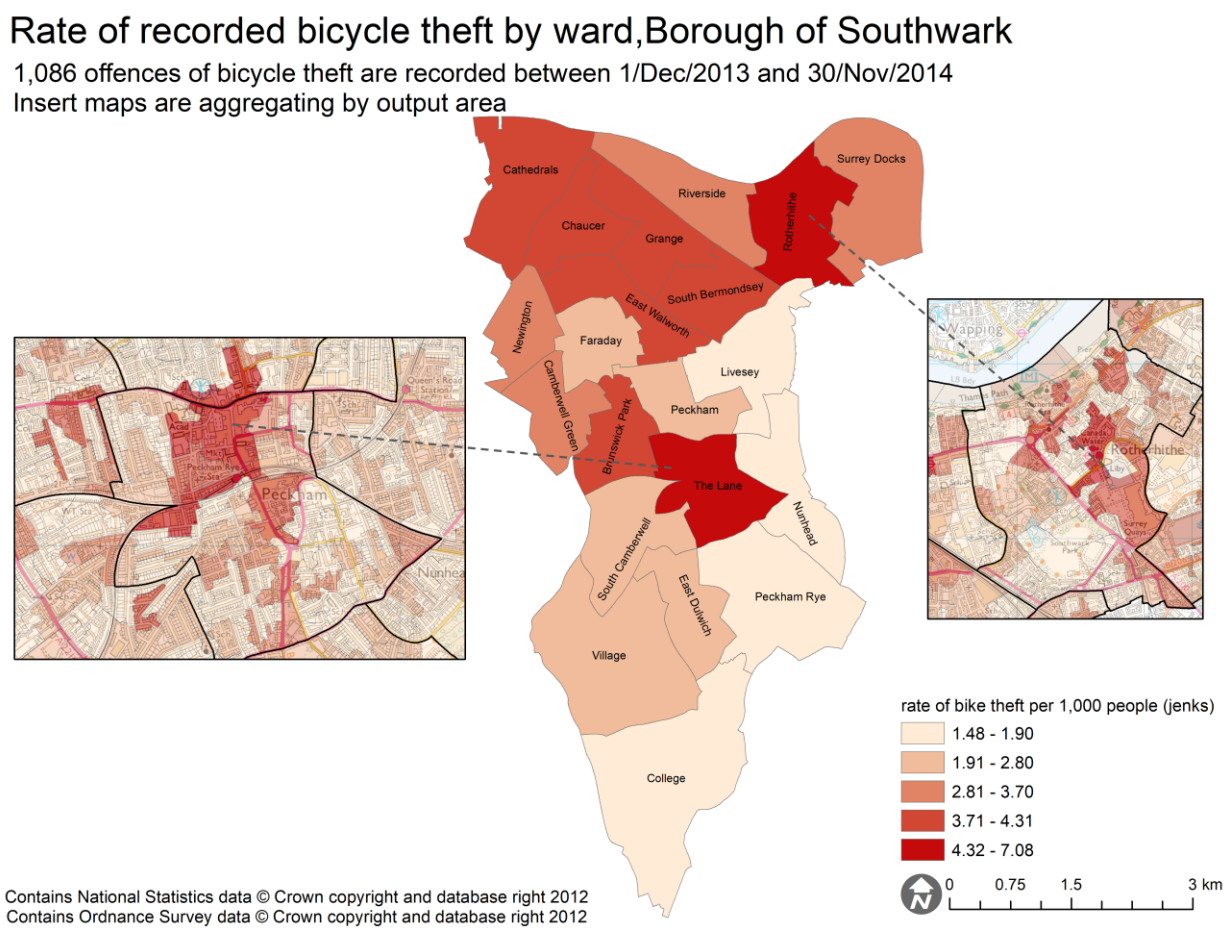
Developing a thematic map that shows crime rates rather than just the count of crimes can better assess risk issues. This is more useful for deciding which ward to focus on for crime prevention programs. To calculate crime rate for bike theft, we adopt Adjusted Population proposed by Mburu and Helbich (2016) as our denominator for bike theft rate, which represents the number of people on the streets who might have their bikes stolen. It is calculated as:

$$Adjusted\ Population\ =\ Workday\ Pop.\times\ \frac{60}{168}\ +Usual\ Residents\ Pop.\times\ \frac{108}{168} \qquad (1)$$

Workday population and usual residents population data are obtained from 2011 Census (Nomis, 2015; Office for National Statistics, 2017), the workday population includes the usual residents plus workers coming into the area minus workers leaving the area. These figures are weighted based on the hours they're likely to spend in the area. It's assumed workers are in the area from 7am to 7pm or from 8am to 8pm on weekdays (Blyton, 2014; Stults and Hasbrouck, 2015), and there are 168 hours in total for each week.

Figure 9 presents a thematic map that displays the bike theft crime rate per 1,000 individuals at risk within various wards in the Borough of Southwark, with deeper colours representing an elevated threat of theft. Top two wards with the highest risk levels are featured in greater details through inset maps by output area, showing the risk is pronounced at certain smaller geographic locations including the recreation place in Peckham and the train station as identified earlier. This suggests that there are specific areas within the wards that have pronounced rates of bike theft, a part from higher foot traffic, there might be repeated victimization in those specific bicycle parking spots. A potential strategy for mitigating theft in these hotspots could be using simple signage intervention like ‘thieves, we are watching you’ (Nettle, Nott and Bateson, 2012).

Figure 9. Crime rate map of Southwark by ward



Q5. Limitations

The report uses 2011 boundary data, incorporating several changes from previous census data, which may not fully reflect recent boundary adjustments. The analysis might not account for crime continuity across administrative boundaries, as shows partly in Figure 8 that crimes do not stop at the boundary. Only monthly information is available, shorter time frames like time period might reveal more useful insights. Additionally, as Figures 2 (a) and (b) demonstrate, using thematic maps based on political boundaries can introduce aggregation bias, potentially misleading interpretations within each mapped area. The kernal-density estimation map was produced using cell size at 50 with fixed bandwidth equals 300, which could be adjusted to fit scenario better.

Additionally, when it comes to allocating resources and assessing areas of potential concern, more advanced methods such as regression analysis with buffer zones can be applied to pinpoint realistic critical regions more accurately than relying on census divisions. Furthermore, the report fails to classify different levels of risk for various highlighted locations, making it difficult for decision-makers to rank the urgency of these areas.

Reference

1. Transport for London (2007). MAYOR OF LONDON Transport for London Transport for London Annual Report and Statement of Accounts. [online] Available at:https://www.annualreports.com/HostedData/AnnualReportArchive/t/transport-for-london_2007.pdf [Accessed 3 Feb. 2024].
2. Mburu, L.W. and Helbich, M. (2016). Environmental Risk Factors influencing Bicycle Theft: A Spatial Analysis in London, UK. PLOS ONE, 11(9), p.e0163354. doi:https://doi.org/10.1371/journal.pone.0163354.
3. Office for National Statistics (2017): 2011 Census aggregate data. UK Data Service (Edition: February 2017). DOI: http://dx.doi.org/10.5257/census/aggregate-2011-2
4. Nomis (2015). WF01BEW (Location of usual residence and place of work (OA level)) - Nomis - Official Census and Labour Market Statistics. [online] www.nomisweb.co.uk. Available at: https://www.nomisweb.co.uk/census/2011/wf01bew [Accessed 8 Feb. 2024].
5. Stults, B.J. and Hasbrouck, M. (2015). The Effect of Commuting on City-Level Crime Rates. Journal of Quantitative Criminology, 31(2), pp.331–350. doi:https://doi.org/10.1007/s10940-015-9251-z.
6. Blyton, P. (2014). Changes in Working Time (Routledge Revivals). Routledge.
7. Nettle, D., Nott, K. and Bateson, M. (2012). 'Cycle Thieves, We Are Watching You': Impact of a Simple Signage Intervention against Bicycle Theft. PLoS ONE, [online] 7(12), p.e51738. doi:https://doi.org/10.1371/journal.pone.0051738.